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Fostering digital equity: evaluating impact of digital literacy training on internet outcomes in rural marginalised communities in India

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ABSTRACT

Digital literacy (DL) training improves the participants' digital skills and engagements and achieves desired online benefits and opportunities. However, there remains a gap in understanding whether the acquired skills effectively translate into tangible outcomes. This study explores how digital literacy training programs (DLTPs) serve as solutions to mitigate the impact of the digital divide, particularly among marginalised rural populations. Using the corresponding fields' model, we explored the internet outcomes achieved post-training, as achieving meaningful outcomes is crucial in the digital age. In-depth interviews of 60 beneficiaries of the PMGDISHA programme, a DLTP run by the Government of India, were conducted, and thematic analysis was performed to identify the internet outcomes based on Helsper's corresponding fields (2012). The findings demonstrate that training significantly impacts digital engagement and internet outcomes across online fields. However, discrepancies in internet outcomes persist among participants, and these disparities are closely linked to participants' socio-demographic and economic characteristics. To create a more equitable digital landscape, digital inclusion initiatives must consider these elements and tailor their approaches accordingly. These findings significantly impact policymakers, managers, practitioners, trainers, and researchers in bridging the digital divide and creating a more inclusive digital society.

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Introduction

In today's digital age, access to the internet has become increasingly vital for individuals to fully participate in society, access information, and engage in various opportunities (Kligler-Vilenchik & Literat, 2020). However, many global populations, particularly marginalised communities, still face barriers to accessing and utilising digital technologies (Lythreatis et al., 2022a). The digital divide raises concerns about social inequality and exclusion in the digital realm (Calderón Gómez, 2021). Addressing the digital divide is not only imperative for promoting social equity but also aligns with the principles of sustainable development (Ye & Yang, 2020) and Sustainable Development Goals' targets delineated by the United Nations such as 4.b (Increasing proportion of youth and adults with ICT skills), 5.b (enhancing use of ICTs to promote empowerment of women) and 9.c (increasing access to ICTs and providing universal and affordable access to the Internet in least developed countries).

Digital literacy (DL) is crucial in mitigating the digital divide by empowering individuals with the knowledge and skills to effectively navigate and utilise digital technologies (Tinmaz et al., 2023). It enables

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individuals to access and use online resources, including educational materials, job opportunities, healthcare information, and government services (Reddy et al., 2022). Lack of DL leads to digital exclusion, which can eventually lead to social isolation (Martin et al., 2016; Warren, 2007), limited opportunities for collaboration and networking (Resta & Laferrière, 2015), and hindered civic engagement (Bélanger & Carter, 2006), barriers to entry in the job market and may be limited to low-skilled, low-wage positions (Lee & Clarke, 2019). Further, this creates educational inequalities and further widens the digital divide among students. Moreover, individuals with low levels of DL are more susceptible to online threats, scams, cyberbullying, or privacy breaches (Susilo et al., 2020). This can have negative impacts on their digital experiences and overall well-being.

To address these challenges, digital literacy training programs (DLTPs) have emerged as a pivotal strategy to bridge the digital divide and empower marginalised populations. These programmes aim to provide individuals with the necessary skills and knowledge to navigate the digital landscape, access online resources, and participate in the digital economy (Choudhary & Bansal, 2022). While DL training holds promise, it is essential to evaluate its effectiveness and understand the outcomes achieved by marginalised populations even after the completion of such programmes.

This study attempts to understand how DLTPs serve as a solution to mitigate the impact of the digital divide, particularly among marginalised rural populations. This study employed the corresponding fields' model to gain insights into the internet outcomes achieved by individuals post-training (Helsper, 2012) as a framework. The model recognises the importance of achieving meaningful outcomes in the digital age and highlights the interconnectedness between offline and digital exclusion.

The Government of India-led DLTP was chosen as the focus of this research because it aims to empower citizens residing in rural areas through comprehensive training that enables them to effectively utilise computer or digital access devices such as tablets and smartphones. By providing these training opportunities, the scheme seeks to bridge the digital divide and empower individuals in rural areas to harness the benefits of digital technologies for personal and professional development. Therefore, by examining the outcomes achieved by individuals who received training through this programme, the study aimed to provide insights into the effectiveness of DLTPs in empowering individuals and bridging the digital divide.

Research gap and motivation

Over the past decade, the field of digital inequality has evolved from a focus on internet access to encompassing internet skills, usage patterns, and outcomes (Ferreira et al., 2021; Hargittai, 2010, Lythreatis et al., 2022b; Scheerder et al., 2017; van Deursen & van Dijk, 2011). DLTPs have been recognised by researchers as a valuable instrument for addressing the digital divide. However, while current research has explored the outcomes of internet use, there is a noticeable gap in understanding the effectiveness of such DLTPs in shaping the internet outcomes achieved by trainees (Julien et al., 2022). In this study, we aim to address this gap by evaluating the impact of DLTPs on the type of internet outcomes trainees achieve. By investigating the extent to which DLTPs contribute to meaningful internet outcomes, this research aims to shed light on the effectiveness of these programmes in mitigating digital inequality and, consequently, their role in addressing the third level digital divide. The third level of the digital divide refers to the disparities in outcomes that arise from differences in access to and utilisation of technology. Unlike the first level, which focuses on access to hardware and connectivity, and the second level, which involves skills and knowledge in using technology, the third level emphasises the consequences and impacts of technology use. This includes variations in the benefits, opportunities, and outcomes that individuals or groups

experience as a result of their engagement with digital technologies (van Deursen & Helsper, 2015b).

Research question

The following research questions guide the study:

(1) What outcomes of Internet use are achieved by trainees post-training?
(2) Do participant demographics explain disparities in Internet outcomes among trainees following their training programme?

Contribution of the study

This study significantly contributes to our understanding of Digital Literacy Training Programs (DLTPs) by delving into their effectiveness in shaping post-training Internet outcomes, with a particular focus on marginalised populations. By empirically examining the outcomes achieved by trainees following DL training, the research offers valuable insights into the tangible impact of these programmes on digital inclusion and Internet utilisation. Moreover, it underscores the critical role of socio-demographic elements to training-specific elements such as programme focus, instructor expertise, curriculum content, training duration, intensity, and pedagogical approaches. Through evidence-based policy recommendations, this study advocates for comprehensive, sustained support and collaborative efforts with marginalised communities to enhance DLTPs and promote more favourable Internet outcomes for those they serve. Furthermore, it highlights the need for further exploration of diverse training methodologies and a deeper understanding of participant demographics to improve these programmes continually.

In the following sections, we will delve into the methodology employed, present the study's findings, and discuss the implications for policymakers, practitioners, trainers, and researchers. Ultimately, this study strives to shed light on the complex dynamics of digital exclusion and contribute to the ongoing discourse on bridging the digital divide and fostering a more inclusive digital society.

Theoretical background

Corresponding fields model

Several studies on the impact of the internet tend to focus on one specific outcome rather than looking at the broader impact of the internet on individuals or society. For example, a Luxembourg household survey (Pénard & Poussing, 2010) examined the relationship between internet use and social capital, specifically the formation and maintenance of social ties, finding that internet use is positively associated with the construction of online and offline social relations. Similarly, Sylvester and McGlynn (2010) investigated the relationship between Internet access, political participation, and geographic location in the United States. They found that individuals with greater access to the internet are more likely to engage in various forms of political participation, such as voting, contacting elected officials, and participating in political discussions. Such studies, which often focus on one particular outcome, use varying terminology to describe outcomes, which makes it challenging to compare and synthesise findings across studies (Blank & Groselj, 2014). Thus, as suggested by Scheerder et al. (2020), a theory-driven approach and a classification of possible outcomes are necessary to provide a more comprehensive understanding of how existing inequalities are reflected in internet outcomes. In this line, Helsper (2012) proposed

the corresponding fields model, which suggests that social inclusion interacts with the fields of digital inclusion. The model posits that offline and online resources are interdependent, and the chance to be included online is reflected by the offline resources one has. The model identifies four fields corresponding to online and offline resources: economic, social, cultural, and personal.

- Economic outcomes are related to an individual's income, employment, education, and property and can include activities like investing online, buying and selling products online, or finding a new job through online platforms.
- Social outcomes are related to an individual's social networks and can include making new friends online, participating in discussions about political or societal issues, or joining clubs or societies online. Social outcomes can also be limited to looking up information about government services online.
- Cultural outcomes are related to an individual's identity and sense of belonging and can include activities related to spirituality, religion, or learning about different ethnicities. Cultural outcomes can also include learning about parenting or childhood development.
- Personal outcomes are related to an individual's health, leisure, and self-actualisation. These can include new ways of entertainment or spending spare time, such as listening to music or watching TV shows via online streaming services or tracking one's movement through applications. Personal outcomes can also include finding new offline events to attend, consulting others about interests, or solving problems related to one's interests.

Overall, the corresponding fields' model provides a framework for understanding how people may benefit differentially from Internet use based on their access, skills and usage and the particular field of outcomes being considered.

Socio-cultural context: internet use and marginalised populations

Marginalised communities refer to those systematically excluded from active participation in mainstream social, economic, educational, and cultural spheres. This exclusion in the context of internet use extends to various groups, encompassing but not limited to those marginalised based on factors such as race, gender identity, sexual orientation, age, physical ability, language, and immigration status (Sevelius et al., 2020). These groups face digital illiteracy and systemic barriers to accessing and participating in the digital world. Empowering them with digital skills and improved internet access is crucial to reducing the digital divide and promoting inclusivity (Correa & Pavez, 2016; Mehra et al., 2004; Ortiz et al., 2019). Internet is identified as a powerful tool for marginalised communities, allowing them to access information that is otherwise difficult to obtain, build social networks with people who share similar experiences and engage in political and social activities that can lead to positive change (Mehra et al., 2004; Mitra, 2004). In his study, Omari (2018) examined the use of digital access amongst marginalised communities from a democratic and internet governance perspective, finding that marginalised communities increasingly rely on digital technologies. Yet, poverty, limited educational opportunities, and lack of access to technology infrastructure can make it difficult for individuals to access and use digital technologies effectively. To accomplish that, digital access is necessary for democratic participation and economic development and calls for more significant efforts to improve digital access (Dlodlo, 2009; Irizarry et al., 2002; Omari, 2018). Moreover, internet use in marginalised communities is associated with greater political participation and a stronger sense of political efficacy (Nemer & Tsikerdekis, 2017). In addition, Mitra (2004) found that women use the internet to access information and resources not available to them offline, such as legal advice and job listings.

Digital literacy training and internet outcomes

Literature highlights the changes in individuals' online activities and habits due to DLTP participation (Matli & Ngoepe, 2020; Michail & Stamatios, 2008; Patankar et al., 2017). This may include increased awareness of the potential risks and benefits of internet use, enhanced ability to access and utilise online resources and improvements in online safety practices (Nedungadi et al., 2018). Such training programmes aim to empower individuals to become more confident and competent digital citizens, able to navigate the internet effectively and responsibly (Choudhary & Bansal, 2022). However, the literature on internet outcomes highlights that not all individuals benefit equally from new digital technologies (van Deursen & Helsper, 2015a; Helsper et al., 2015). Participant demographics shape not only internet usage, digital proficiency, and digital engagement but also the outcomes of online practices (Hargittai, 2002, 2010; Hargittai & Hinnant, 2008; van Deursen & van Dijk, 2011; van Dijk, 2005; Zillien & Hargittai, 2009). In their study (Scheerder et al., 2020), identified that as the use of the internet increases, people with higher levels of education are more likely to learn and benefit from this usage, thus further widening the gap between groups with different socioeconomic backgrounds. Privileged individuals tend to use the internet more effectively and reap more significant benefits, be they economic, social, or health-related, compared to less-privileged users (Bonfadelli, 2002; Zillien & Hargittai, 2009). This highlights the potential for the digital divide to exacerbate existing social inequalities based on educational attainment (Ragnedda & Muschert, 2013; Robinson et al., 2020). However, these differences in internet outcomes remain after basic DL training – and have been under-researched till now. Evaluating the long-term impact of DLTPs can help determine programmes' effectiveness on participants' internet usage behaviour.

Methodology

Study design

This exploratory study adopts a qualitative approach recognising actions enmeshed with the broader socioeconomic environment. This approach was chosen because of the need to understand the phenomena of initiatives addressing the digital divide, which is complex and influenced by diverse cultural and socioeconomic experiences. The field of inequality research has predominantly relied on quantitative analyses, which offer limited explanatory power when it comes to understanding the underlying reasons for disparities (Scheerder et al., 2017). While these quantitative methods can identify which segments of the population are better served by the internet, they fail to provide insights into the underlying determinants of these disparities (Hacker & Mason, 2003). To address this gap and provide a more comprehensive understanding, this study adopts a qualitative approach that examines the social context in which internet use occurs, explicitly focusing on marginalised rural communities.

Study context

The study was conducted in rural areas across eight districts of Rajasthan, a state in northwestern India. We took Pradhan Mantri Gramin Digital Saksharta Abhiyaan (PMGDISHA) as a case study to understand the effectiveness of DLTPs in empowering marginalised rural communities in India. Launched in 2017, PMGDISHA is a government-run digital literacy training programme that aims to make individuals digitally literate, empowering them to access and utilise digital technologies for active engagement in governance and nation-building processes (for more details, see Appendix). By equipping citizens with digital skills, PMGDISHA intends to enable marginalised and rural communities to participate in nation-building and contribute to the government's vision of a digitally empowered society.

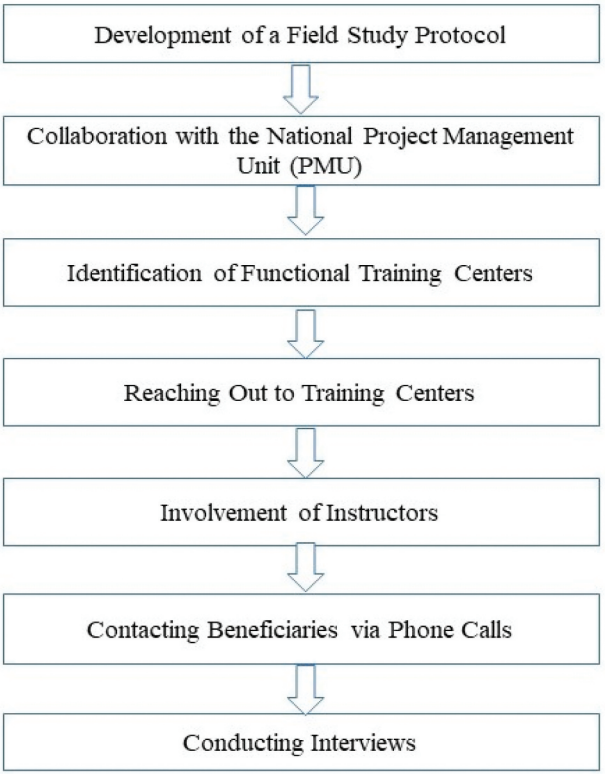


Figure 1. Field study protocol.

Field study protocol

A field study protocol (see Figure 1 was crafted before the fieldwork to maintain consistency in data collection. This protocol not only established a structured framework for data gathering but also possessed the flexibility to be tailored to specific circumstances. To establish contact with the programme's beneficiaries, the researchers collaborated with the national Project Management Unit (PMU), responsible for overseeing PMGDISHA. They partnered with the PMU manager to pinpoint the operational training centres and their respective beneficiaries. Subsequently, outreach was conducted with a total of 20 training centres. The instructors at these centres played a crucial role in facilitating connections with the beneficiaries. A list provided by the instructors guided the researchers in making phone calls to the beneficiaries, and those who agreed to participate in interviews were invited to the centres where the interviews were conducted.

Data collection procedure

Before each interview, participants received a thorough briefing on the research's goals and methodologies, and their informed consent was duly secured. Interviews were conducted in Hindi for some participants, while others were conducted in various local dialects, with a local assistant accompanying the researcher to capture contextual and specific details. Interviews lasted between 30–40 minutes. They were recorded and transcribed verbatim to identify internet outcomes. Also, before commencing the interviews, the researchers created a checklist of outcomes. This checklist outlined the internet outcomes, derived from an inventory of potential outcomes developed and categorised according to Helsper's (2012) framework encompassing economic,

Table 1. Demographic profile of participating individuals.

	N	%
Gender		
Male	27	45
Female	33	55
Age		
14–30	31	51.7
31–45	21	35
46–60	8	13.3
Education		
High education attainment	30	50
Low education attainment	30	50
Occupation		
Students	25	41.7
Employees (Government)	2	3.3
Employees (Private)	5	8.3
Self-employed (Business)	16	26.7
Homemakers	12	20
Total	60	100

personal, cultural, and social fields. Each participant's internet outcomes achievement was marked on the list during the interview. Participants were not requested to mark the outcomes to avoid potential self-reporting bias. This approach encompassed both deductive and inductive elements. Initially, outcomes were derived deductively from the predefined fields model. However, participants were also encouraged to articulate any additional outcomes they experienced.

Sampling and participants' profile

Participants were chosen through a purposive sampling method, specifically targeting individuals who had undergone training within one year of the programme's conclusion. The study participants comprised individuals who were beneficiaries of the PMGDISHA programme (see Table 1). They were 14 to 60 years old, representing diverse occupations, including students, employees (both in government and private sectors), self-employed individuals (engaged in business), and homemakers. Gender diversity was considered to ensure a well-rounded representation of the beneficiary population, including male and female participants. The term 'lower education attainment' (LEA) was assigned to individuals with educational qualifications at or below the intermediate level, while 'higher education attainment' (HEA) was used for respondents holding a graduate degree or higher. These categorisations allowed the study to comprehensively explore and analyse how age, education levels, and gender might impact internet usage patterns and outcomes among PMGDISHA beneficiaries.

Data analysis

All interviews underwent verbatim transcription and were integrated into the analysis. We employed thematic analysis methodology to organise the transcribed text according to Helsper's (2012) corresponding fields. These interview questions are directly aligned with these fields. Subsequently, we queried the data to identify emerging patterns and responses within these fields. A deductive-inductive approach, combining two methods, was utilised. Initially, outcomes were primarily derived from the corresponding fields model (deductive). Subsequently, participants were encouraged to express their outcomes, which were then analysed and, when applicable, categorised within one of the corresponding fields (inductive). This approach allowed for a structured framework while also accommodating participant-generated insights. In their study, Scheerder et al. (2019) followed a similar procedure when identifying the negative outcomes of Internet use within different educational backgrounds.

In addition to the primary researcher, a secondary coder (co-author) was enlisted to independently analyse a subsample of transcripts, comprising 10% of the total dataset. This step was undertaken to ensure the reliability of the coding process and to assess intercoder reliability. The calculated Cohen's kappa value of .82 indicated a high level of agreement between the two coders, affirming the robustness and consistency of the coding process. Finally, the primary research objectives were achieved, and core findings corresponding to the four designated fields were successfully identified.

Findings

RQ1 outcomes of internet use achieved by trainees

In this section, the outcomes of internet use achieved post-training are discussed. The outcomes are categorised under the economic, cultural, social, and personal fields to address the internet outcomes using the respective fields' model classification.

Economic

The economic field is one area where the outcomes achieved by participants are most noticeable. Participants tend to have access to more types of economic benefits post-DL training.

Economic – income. Post-training participants have become accustomed to buying and selling online. It is reflected in their motivations and behaviours. They tend to prioritise price when shopping online, often looking for the best deals to save money and value the convenience and time-saving aspect of services available online.

> Recently, I bought a water cooler at a 40% discount . . . it was delivered directly to our home. Buying items online can be convenient, especially for larger items that can be difficult to transport. (Male, 28 years, LEA, self-employed electrician)

Moreover, the DL training helped participants transform how they conduct financial transactions. Participants have moved away from traditional methods of cash transactions and adopted mobile payment applications like Paytm, digital wallets, and Unified Payment Interface (UPI). They learned to make transactions faster, safer, and more convenient. A few participants saved money on cashback offers that provided discounts on future transactions instead of an immediate refund. This way, they have saved money in the long run by reducing the cost of future transactions.

> I learned about the benefits of using digital payment apps during training . . . they are easy to use, and to my surprise, every month, I can save some money now, either on discount or cashback offers. (Female, 32 years, LEA, homemaker)

Individuals with self-run businesses are more likely to achieve commerce-related outcomes than employed people, which may be because their prospects for business expansion have increased after their DL training. They have more opportunities and flexibility to engage in online commerce-related activities.

> Smartphone use has improved my skills in making rugs by referring to YouTube. Recently, I have started marketing the material on Facebook, which has attracted more visitors to our doorstep . . . (Female, 31 years, HEA, self-employed artisan)

Furthermore, in the case of small-scale business vendors, their internet usage has pushed them to access better online entrepreneurship opportunities, such as starting an online store, offering services, or launching a digital product. This has provided them with an additional income stream and increased financial stability.

> After the training, I started to use WhatsApp Business to create a catalogue to send to customers living in different places. Reconnecting and reaching out to customers is easier than physically travelling to cities to sell. (Male, 28 years, HEA, self-employed artisan)

Using WhatsApp Business as a communication platform has enabled many small-scale (mainly home-based) businesses to connect with their customers, promote their products and services, and manage their business operations. It has empowered them to promote their products and services by creating and sharing a business profile with their customers. Moreover, the customers can send messages, ask questions, and place orders, which the business owner can manage through the app.

Economic – finance. Findings prove that DL training helped individuals improve their financial literacy by accessing online resources such as budgeting tools, financial calculators, and investment information. This ultimately improved financial decision-making and increased financial stability. Moreover, participants can access more online banking and financial services, such as online bill pay, mobile deposits, and investment platforms. Participants confirmed greater convenience and flexibility in managing finances post-training. Moreover, DL training developed competencies that enabled individuals to access online marketplaces, such as e-commerce websites, peer-to-peer lending platforms, and crowdfunding sites. This has provided alternative sources of financing and investment opportunities.

> During training, I learned to use digital banking, and now I can explore the different money investment offers available with banks. And I've slowly started to save and invest money now. (Male, 34 years, HEA, private employee, assistant)

However, it is observed that participants with higher education attainment (HEA) and employed status are more likely to engage in new types of investments, including internet-based ones, such as opening fixed deposit accounts or conventional forms of investment, which generally require an online search for information-seeking.

Furthermore, participants now know how to protect themselves against financial fraud after learning about online security measures and identifying potential scams in DL training. They feel competent to prevent financial losses and maintain financial security.

> PMGDISHA helped me understand common scams that fraudsters use to steal money. I know how to protect my personal and financial information by keeping strong passwords and turning on encryption. (19 years, male, LEA, student)

Economic – employment. Work-related internet use is an essential part of daily work. The internet is used more for supportive tasks or administrative purposes. Findings indicate that after the training, participants started to rely heavily on the internet to perform their jobs and use it to improve their work continuously. Additionally, irrespective of the job profile (ICT-based or other), it is clear that no job would last longer without the internet. Thus, participants are generally more satisfied with the work-related outcomes. They use the internet to improve their work continuously, integrate new software, and find online solutions to make their work more efficient and effective.

> I have switched to the digital payment system after the training. Online transactions allow for more flexible payment options. There are no worries about keeping count of the coins or paper money now. I feel more relieved and tension-free now. (Male, 45 years, LEA, self-employed shopkeeper)

Additionally, labour-related economic outcomes, such as income and job opportunities, are most likely achieved among younger people. Unlocking potential with Google Assistant, a woman tailor says,

> During the training, I came across Google Assistant, which has set almost everything in my life. It was astonishing that a voice was teaching me everything from business strategies and stitching techniques to listening to music and other entertainment purposes. I learnt to make masks during. COVID-19. (Female, 21 years, LEA, self-employed tailor)

The reflective attitude of employed participants towards their work is also reflected in their approach to career development. They tend to be more proactive in adopting digital methods to organise work, search for more opportunities, and continuously update their online presence.

> I can now quickly calculate the daily sales and profit generated through the business using the calculator feature in the smartphone . . . I've also started to maintain inventory on Excel on my PC. (Female, 31 years, HEA, self-employed artisan)

DL training has empowered participants to access professional networking opportunities like LinkedIn and other social media networks. This has expanded their professional network, enabled them to connect with potential employers, and enabled them to learn about job opportunities. Their post-training internet use has evolved; now, they are competent to expand the pool of possible job opportunities and can apply for jobs more efficiently.

Economic – education. DL training has proved to help expand participants' knowledge and skills, making them more competitive in the job market. Now, participants often see their work as an opportunity to learn and grow, and they actively seek out courses or workshops that can help them improve their skills and knowledge. They are often proactive and see it as a way to invest in their career development.

> Earlier, I used to experience challenges while teaching because textbooks do not always offer enough or different explanations, but now the internet has fixed this issue. (Male, 48 years, HEA, self-employed tutor)

Participants have learned to access online education and training resources, including Massive Open Online Courses (MOOCs), tutorials, and webinars. Now, they are expanding their knowledge and skills, making themselves more competent for the job market. Such educational outcomes are more relevant to the student group.

> . . .I have registered and joined online classes for government services preparation. (Male, 23 years, HEA, student)

> Using the internet, I learnt about several educational courses and the opportunity for graduation from distance learning. Now, I am pursuing my graduation in distance learning mode and am in 2nd year. (Female, 19 years, LEA, student)

On the other hand, other participants often learn through informal channels. Such participants may not always see the relevance or benefit of this way of learning, but they subconsciously educate themselves through these channels.

> PMGDISHA taught me how to use a smartphone effectively. Post-training, I picked up skills like tailoring and embroidery through YouTube videos. Mobile is a blessing in disguise for me. It is like a genie that helped me become financially independent. (Female, 37 years, LEA, self-employed tailor)

The different approaches to work-related courses or tutorial videos highlight the different attitudes towards personal development.

Social

Social- informal. Based on the provided information, there are specific patterns and trends in the relationship between living situation, relationship status, age, and gender and the likelihood of achieving social outcomes related to internet use. All types of people – whether living alone or with others, such as roommates or family members – experience social outcomes of internet use, which may be due to the increased opportunities for social connection and interaction that come with improved digital access and

internet connectivity infrastructures. Participants seem to use the internet to maintain social contact, with most indicating that online communication facilitates quick and frequent opportunities to stay in touch with friends and family. This often results in the strengthening of current connections. Participants also mention that online contact helps them stay up-to-date with the lives of their loved ones and facilitates offline appointments. However, most participants agreed that online contact could never replace the value of face-to-face interactions.

> Earlier, I used to spend my time online watching videos or listening to music, but now I can find and contact people who are important to me using social media platforms. Since most of them are family members, we feel more connected and share everything. (Male, 46 years, LEA, self-employed plumber)

> My husband works outside. . . I talk to him over video call every day. Distance does not feel like a distance because of the Internet. (Female, 27 years, LEA, homemaker)

Considering the statements, it may not be wrong to contend that individuals living in far-off places are likelier to experience social outcomes related to internet use. It may reflect that lonely individuals – widowed, elderly or divorced – may be more motivated to seek social connections and support online (Bansal & Choudhary, 2023).

Social – formal civic participation. Participants revealed how the internet had broadened their horizons and helped them become more aware of different cultures and issues worldwide. This can lead to a greater appreciation of diversity and a more informed understanding of global issues. Exposure to different cultures and ways of life may also develop a more open-minded and tolerant worldview. One may learn to appreciate the similarities and differences between cultures and gain a deeper understanding of the challenges and opportunities faced by people in different parts of the world.

> Internet use has broadened my horizon as I am more aware of the different issues in other parts of the world . . . what socio-cultural practices other countries follow . . . what festivals they celebrate . . . I feel well aware now. (Female, 37 years, LEA, homemaker)

> With internet use, I feel like contributing to social issues . . . For instance, recently, I came across an online petition request against animal cruelty on my email account. I would not have replied to such petitions if I had not learnt how to reply to emails in training. (Male, 26 years, HEA, private employee, clerk)

Social – political. Post-training, participants have started to access and evaluate political information, including news articles, policy proposals, and political commentary. They are now actively involved in political discussions through online forums, social media, and other digital platforms. This has ultimately influenced their participation in political processes, such as voting, connecting elected officials, and attending public meetings. Moreover, participants now understand how digital media can be used to spread misinformation and propaganda. They can better recognise and counteract these tactics and engage more critically with online political content. This helps them make informed decisions and engage more meaningfully in political discourse. This can enable them to share their perspectives, learn from others, and contribute to public debate.

> Before learning to use the internet, I had minimal knowledge of political affairs at local and national levels, but now I have access to every piece of information on my mobile. . .I am more aware of my city's local leaders and regularly follow their updates on Facebook and YouTube. (Female, 38 years, LEA, widow)

Cultural – belonging and identity

DL training has helped individuals learn how to use social media and other online tools to connect with others who share their interests, thus inculcating a sense of community and belonging, even for individuals with limited in-person socialisation opportunities.

> After completing the training program, I learned to chat with people… upload my pictures as statuses on Facebook and WhatsApp. We make meeting plans with friends on WhatsApp groups and coordinate accordingly … I feel more connected than now. (Female, 39 years, LEA, homemaker)

Women often seek information on parenting and upbringing to get guidance and solutions for their challenges, such as sleep problems in babies or breastfeeding toddlers. This information helps them with tips and tricks, leading to convenience and time savings as positive outcomes.

> I have found a few Facebook pages providing useful information about dealing with children and their issues. Since then, I have regularly read updates about children and their requirements and tried to apply that knowledge in practice. (Female, 23 years, LEA, homemaker)

Overall, participants expressed how their internet usage is evolving. They can now access cultural resources online, including art, music, literature, and historical archives. This has promoted a greater understanding and appreciation of cultural heritage and fostered a sense of belonging to cultural communities. Participants convey how they explore and learn about other cultures through online resources, including language learning tools, virtual tours, and social media connections. This is reflected in their changing perception of promoting the culture through work, developing a greater understanding and appreciation of diversity, and promoting cross-cultural socialisation. Moreover, the internet can provide a platform for cultural expression and encourage sharing of cultural traditions and perspectives. Participants share how they create and share cultural content online, such as writing, art, music, and photography.

> I am a weaver. I love to weave the fabric of my culture, and I am responsible for keeping it alive, which I am now doing through my smartphone. (Male, 32 years, LEA, self-employed weaver)

This reflects how internet use has led to a more enriched cultural life and a greater sense of belonging and identity in cultural communities.

> One Facebook page has enabled me to access more information about my ethnic group. I have learned many unknown facts about my community that I did not realise earlier. (Female, 22 years, HEA, student)

Personal

The internet helps access information, resources, and personal growth and development opportunities. Using online resources, participants acquire new knowledge and skills to help them achieve their goals and reach their full potential.

Personal – self-actualisation

Evaluating their post-training internet use behaviour, participants reported increased skills and knowledge needed to navigate technology effectively. As learners become more proficient in using digital tools, they may feel more confident in their abilities, which leads to increased self-esteem.

> I am illiterate and cannot read and write, but because of my mobile, I am no more handicapped. I can do everything that I was unable to do earlier. (Female, 57 years, LEA, homemaker)

> Post-training, and with time, I've learnt how to receive calls, share photos, and search videos on YouTube- all by identifying and recalling the icons of these apps. I now feel more empowered as a woman. (33 years, female, LEA, homemaker)

Often, DL training involves learning how to troubleshoot common problems that arise when using technology. As learners become more skilled at problem-solving in a digital context, they may apply these skills to other areas of their lives, which can lead to increased self-actualisation. As learners experiment with these tools and explore their potential to

inspire creativity and innovation, they discover new ways of expressing themselves and pursuing their interests. This leads to a deeper understanding of the world around them and helps them make more informed decisions, which can contribute to a sense of self-actualisation.

> I have never been to school and had no mentors who could guide me through the right things or provide me with knowledge, but now I have found one. This device is my mentor. It keeps me informed and shows me the path. (Male, 58 years, LEA, self-employed electrician)

> I know how to change the language in Hindi by going into my phone settings. Language is no longer a barrier; I can explore and do whatever I wish to do . . . all by myself. (Female, 33 years, LEA, homemaker)

DL training can provide learners with the skills and knowledge to navigate the digital world confidently and creatively, contributing to their overall understanding of self-actualisation.

Personal – health

In health, participants indicated how they search for medical or health-related information online through online apps or websites. However, participants with low educational attainment are not found to rely much on online information for health purposes.

> In training, I learned to explore and download different types of apps. One app I downloaded is related to health. I've been continually following the heath chart that it suggests . . . . I've noticed a lot more improvement in my health. (Male, 43 years, HEA, private employee-clerk)

In addition to health apps, HEA participants more frequently mention using the internet for mental health purposes, such as finding information about mental health issues and online self-help resources, such as meditation or mindfulness apps. Again, participants with low educational attainment did not mention using the internet for mental health purposes, and when they do, it is often for basic information gathering rather than utilising online resources for ongoing support or treatment.

Participants communicate more effectively with their healthcare providers through email, telehealth, or patient portals. This has helped improve the quality of care and ensure that individuals receive the support they need to manage their health.

> I no longer stand in queues for an appointment with doctors. Now, I seek appointments in advance on the online portal. (Female, 28 years, HEA, teacher)

Moreover, participants with HEA use digital health tools to manage their health. For example, they may learn how to use apps or wearables to track their physical activity, monitor their medication adherence, or manage their chronic conditions.

> I keep track of my health using my smartwatch, as it offers health-related features, such as heart rate monitoring, sleep tracking, and calorie tracking (Male, 22 years, HEA, student)

Personal – leisure

With internet browsing, people have discovered new leisure activities they enjoy online, such as streaming movies, playing games, or participating in virtual communities. All these have expanded their leisure options and provided a source of entertainment and relaxation. Some mobile applications have become increasingly popular on social media platforms, offering a range of benefits and drawbacks depending on how they are used and consumed. While many users enjoy reels as an entertaining way to engage with content, others have raised concerns about their impact on mental health and the dominance of algorithm-driven content on the platform.

> Some mobile applications like Facebook, YouTube, and Instagram are entertaining ways to discover new content and pass the time. (Female, 23 years, LEA, student)

> I see my children scrolling through reels without realising how much time has passed . . . (Female, 34 years, LEA, homemaker)

RQ2 participant demographics affecting the internet outcome achievement

Gender and education

The study's investigation into identifying the participant demographics affecting the internet outcome achievement yielded several noteworthy conclusions: First, engagement in a specific online activity does not necessarily result in tangible offline outcomes.

> I've been using social media to connect with people but haven't yet found a trustworthy, lifelong connection. (Female, 33 years, HEA, government employee, clerk)

> Identifying genuine buyers is challenging due to encounters with individuals using vulgar language, making me cautious about relying too much on online marketing (Female, 26 years, HEA, artisan)

This statement highlights the complex nature of online interactions, where the vastness of the digital landscape often offers numerous connections but may sometimes lack the depth and trust that come with long-term relationships. Women and gender minorities often encounter gender bias, harassment, and stereotypes in online spaces, which can hinder their ability to benefit from digital networks fully. Educational backgrounds empower people with enhanced online navigation skills, facilitating richer connections in virtual spaces. Conversely, those with lower levels of education may face challenges or difficulties when engaging in virtual environments. Additionally, it is noted that the occupation has a direct bearing on the nature of online social interactions, as professionals in tech-related fields or networking-intensive industries tend to be more adept at leveraging digital connections for personal and career-related gains.

> I can make calls and browse on my smartphone, but my lack of typing-writing skills hampers my ability to support my husband's business and is compounded by my limited education. (Female, 38 years, homemaker)

Age

Second, individuals who successfully attain an outcome in one field of their online activities may not necessarily replicate that success in another area. For instance, the following experience highlights a common hurdle businesses face in the digital era.

> I've built a substantial online social network, but it hasn't translated into attracting customers for my business as expected. I have to advertise my products, visiting shops individually to generate sales.

He further added,

> . . . however, this isn't the case with my younger competitors. They have effectively utilised various digital platforms to enhance visibility and resourcefulness. They demonstrate better knowledge and activity. I can only replicate what I observe and learn from my peers. (Male, 42 years, LEA, self-employed salesperson)

Despite successfully cultivating a sizeable online social network, the gap between online networking and business outcomes underscores the complexities of online marketing and customer acquisition. As a result, several people resort to traditional, offline methods such as visiting shops individually to generate sales, reflecting the need for a holistic and adaptable approach to business promotion that combines both online and offline strategies. Younger people, inherently attuned to digital technologies, often enjoy a competitive edge in navigating the complexities of the online marketplace. Their innate digital literacy and adaptability facilitate rapid learning and adaptation to ever-evolving online trends, ultimately enhancing their profit-making potential. Similarly, those with higher education, especially in technology-related or business fields, may better understand critical aspects of online business, such as e-commerce,

data analysis, and digital marketing. This knowledge empowers them to make informed decisions, identify profitable opportunities, and strategically position their ventures for success.

Education

Third, digital skills play a crucial mediating role in transforming online engagement into tangible, beneficial outcomes. These skills are the gateway to accessing a wealth of information and resources, enabling individuals to make informed decisions and take actions that lead to positive results.

> My internet usage revolved around Facebook and general browsing, but then I discovered how my peers adopted digital banking and online shopping. After receiving the training, I'm more of an online person . . . if I wish to learn more, I seek help from my younger and educated ones. (Female, 59 years, HEA, retired government employee)

The above-quoted statement reflects evolving and transformative internet usage patterns. Initially centred on social media and general browsing, digital engagement increases with time and skills. Several other factors, such as access, awareness, social support, perceived usefulness, education and training, and intergenerational learning, play significant roles in shaping one's digital journey. For instance, older females are less likely to have personal access to smartphones or the internet, suggesting that internet use tends to be dominated by males among older adults.

> I received the training out of sheer curiosity, and I don't even have a personal smartphone. I realised that purchasing one wouldn't make much sense without understanding how to use it effectively. (Female, 56 years, LEA, homemaker, mother)

The woman's statement showcases a remarkable blend of curiosity, pragmatism, purpose, and a commitment to lifelong learning in the context of technology adoption. This pragmatic perspective is particularly noteworthy – she recognised that owning a smartphone would only be meaningful if they could use it effectively. This reflects a deliberate and purposeful mindset, likely driven by specific goals or anticipated benefits associated with smartphone use, such as improving communication or accessing information. Nevertheless, the ramifications of the digital access gap are substantial, contributing to social isolation, economic disadvantages, and disparities in healthcare access. This scenario aligns with the idea that access and ownership of digital devices are only part of the equation when addressing digital exclusion. Digital skills serve as the linchpin in translating online engagement into tangible outcomes.

> I utilise the internet for tasks that don't demand advanced digital skills. I've found my way through the learning process quite skilfully. Whenever I encounter something new, I rely on my social circle and often turn to the younger generation, who can guide me on specific functions. (Male, 56 years, LEA, self-employed shopkeeper)

The statement reflects a conscious choice to engage in online activities that do not demand advanced digital skills, yet the self-learning ability indicates a level of digital adaptability, enabling the learner to navigate the online landscape skilfully. What stands out is their reliance on their social circle and the younger generation for guidance when encountering unfamiliar online functions. This highlights the value of social learning, where knowledge and expertise are shared within a network. Turning to the younger generation for specific functions underscores the intergenerational digital skills and knowledge exchange. By seeking assistance from the social circle and younger family members, he demonstrates resourcefulness in navigating the digital world, even without formal education.

Discussions

This study has delved into the pivotal role of DLTPs in mitigating the challenges posed by the digital divide, especially within marginalised rural communities. It has focused on achieving meaningful internet outcomes, a critical aspect in our increasingly digital age. The third-level digital divide has been a central concept in our exploration, shedding light on the persistent disparities that reappear even in regions where initial access and skills-related inequalities have been minimised. Remarkably, while

access, skills, and positive attitudes towards ICTs are foundational for digital inclusion, they alone cannot comprehensively bridge the divide (Helsper, 2012). It is imperative to conceptualise different inclusion levels to address this, reflecting how individuals engage with technology across various domains such as entertainment, information-seeking, finance, and social interactions (Livingstone & Helsper, 2007; Witte & Mannon, 2010).

The study emphasises that essential digital skills, encompassing operational, information navigation, communication, problem-solving, creative, and safety skills, are the building blocks for translating online interactions into real-world advantages (Ragnedda et al., 2022). DLTPs have emerged as instrumental tools in closing the digital divide, empowering individuals regardless of age, gender, education, or occupation. These programmes facilitate informed decision-making, enhance employability, promote social inclusion, and foster lifelong learning (Choudhary & Bansal, 2022). Moreover, they contribute to gender equality, income generation, and cybersecurity awareness.

However, despite the profound impact of DLTPs, the study has revealed that disparities in internet outcomes persist among participants, primarily due to the participant demographics. Age, gender, education, and occupation influence digital usage and outcomes (Bansal & Choudhary, 2023). Younger individuals with higher education and income levels tend to achieve more diverse and advanced internet outcomes (van Deursen, 2020). Economic elements, including financial resources and employment status, significantly affect internet outcomes, limiting opportunities for those with fewer resources. The study also underscores the role of informal learning and social support networks in bridging digital divides. Individuals often rely on their social circles and younger family members to navigate the digital realm, showcasing resourcefulness in adapting to the digital world, even without formal education (Petrovčič et al., 2022). Indeed, prior research (for instance, Collin et al., 2018; Dimaggio & Garip, 2012; Helsper, 2016) has hinted at the potential influence of network effects on digital engagement and outcomes. As explored by Helsper and Eynon (2010), family dynamics have shown that while children may influence access acquisition and, to some extent, skills development, the actual uses of ICT are more closely linked to the socio-cultural characteristics of parents.

This intricate interplay of socio-demographic and economic elements highlights the need for inclusive government policies and initiatives that address affordability barriers, provide equal access to digital skill development opportunities, and promote digital literacy across diverse populations (A. Scheerder et al., 2017). Tailoring programmes to the unique needs and cultural contexts of the communities served is essential for their effectiveness. Furthermore, this study encourages trainers to adopt a holistic approach beyond imparting digital skills. Collaboration, continuous learning, and improvement are crucial to effective training outcomes.

Finally, the study underscores the importance of future research endeavours to delve deeper into the complex relationship between offline and digital exclusion and the mediating elements influencing internet outcomes for marginalised rural populations. Evaluating the effectiveness of training programmes in addressing offline and digital exclusion and fostering collaborations with marginalised rural communities will enrich our understanding of effective digital literacy interventions (Bansal & Choudhary, 2022). Academic institutions should take the lead in promoting interdisciplinary collaborations to develop comprehensive frameworks and methodologies that inform the design and implementation of impactful digital inclusion strategies.

Limitations and future scope

Generalisability is a concern in qualitative research. While this study provides valuable insights into a wide range of outcomes achieved by training participants, its applicability to larger populations is limited due to its exploratory nature and focus on a specific rural developing country – India. Future researchers can replicate the study in diverse settings and extend the findings of this explorative study through a large sample study. This would allow for a more representative sample and statistical analysis of the relationship between digital engagement and outcomes in different populations. It would also be valuable to conduct cross-cultural studies to examine how these outcomes vary across other socio-

cultural contexts. Additionally, longitudinal studies can provide better insight into the internet outcomes achieved by the trainees over time.

Conclusion

This study has explored the vital role of digital literacy training programs (DLTPs) in mitigating the digital divide, particularly in marginalised rural communities. It highlights the importance of achieving meaningful internet outcomes, a vital aspect of the modern digital age. The third-level digital divide underscores persistent disparities, even when initial access and skills-related gaps are minimised. Essential digital skills, encompassing operational, information navigation, communication, problem-solving, creative, and safety skills, are essential for translating online interactions into real-world advantages. DLTPs empower individuals of all backgrounds, promoting informed choices, employability, social inclusion, and lifelong learning. Informal learning and social support networks are crucial in bridging digital divides. People often turn to their social circles and younger family members to navigate the digital world effectively. However, disparities in internet outcomes persist due to participant demographics. Age, gender, education, and occupation influence digital usage and outcomes. Government education and training policies should adopt a comprehensive approach, encompassing collaboration, continuous learning, and improvement for effective training outcomes. Future research should explore the intricate relationship between offline and digital exclusion and the mediating elements affecting internet outcomes for marginalised rural populations. Collaboration between academic institutions can inform impactful digital inclusion strategies.

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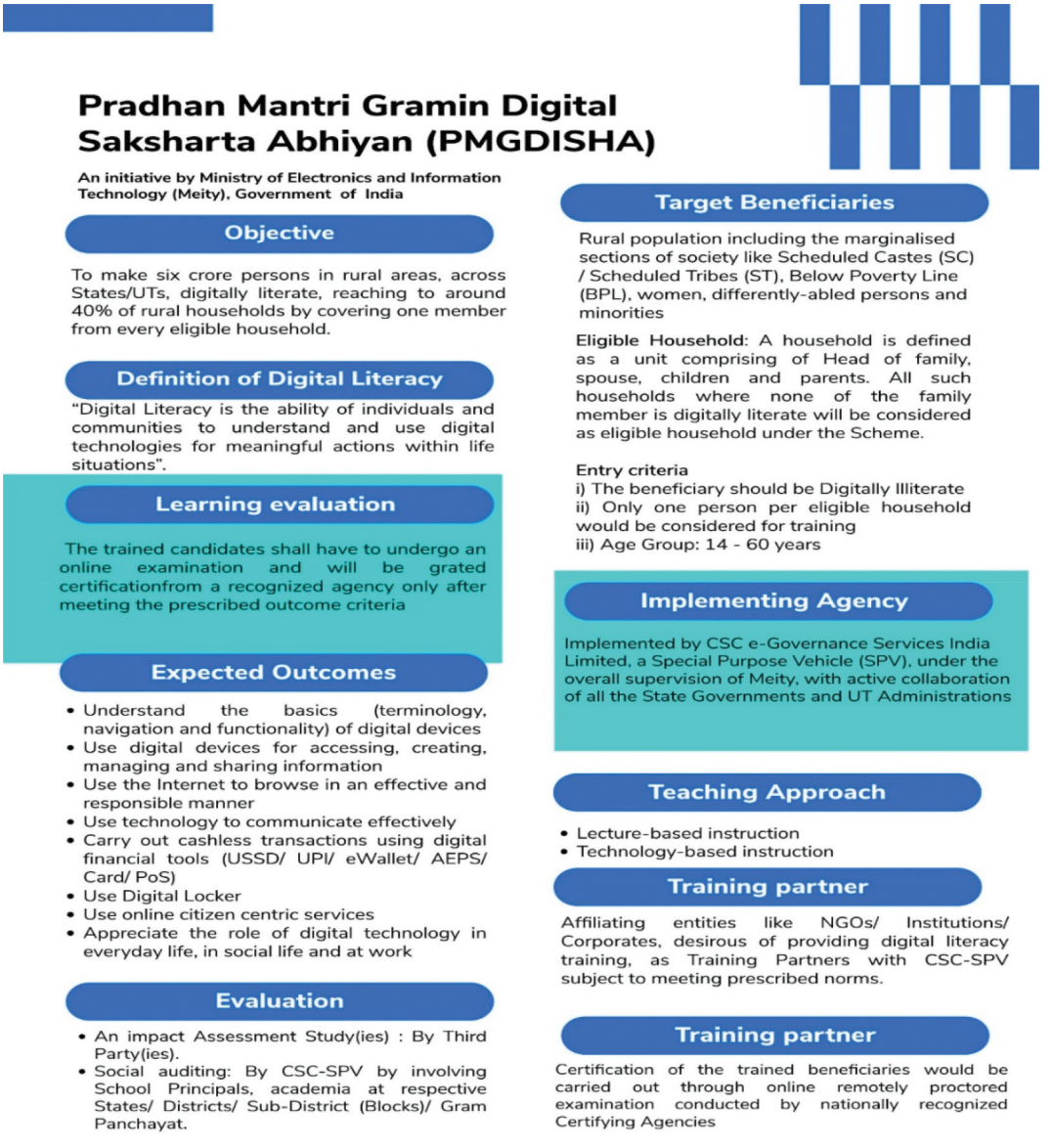
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Appendix: An overview of PMGDISHA



Source: Authors' compilation